Composite inner exponents in shifted binomial compositions

Mathematical research addendum prepared with AI assistance

Project series: Noah's GPT-6 Astra math problems

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Abstract

The affine-power exclusion in the accompanying prime-inner-exponent paper immediately extends its irreducibility conclusion to every inner exponent whose prime divisors exceed the outer prime. Repeated prime factors are allowed. This addendum records the short root-field argument, the exact Lean endpoint, and the verification boundary of the new source. No originality claim is made, and the full fixed-gap Pillai conjecture remains unresolved.

1 Statement and scope

For a nonzero integer k and a rational prime ℓ, let $v_\ell(k)$ denote the exponent of ℓ in $|k|$.

Corollary 1. *Let $p \geq 5$ be prime, and let $m > 1$ be a natural number such that*

$$q \mid m, \quad q \text{ prime} \quad \Longrightarrow \quad q > p.$$

Let $k, A \in \mathbb{Z}$ satisfy

$$|k| > 1, \qquad v_\ell(k) < p \quad \text{for every prime } \ell \mid k.$$

Then $(X^m - A)^p + k$ is irreducible over $\mathbb{Q}$.

Both k and A may have either sign, and $A = 0$ is permitted. There is no assumption $\gcd(m, k) = 1$ and no squarefree condition on m. In particular, primes dividing m or the outer prime p may divide k, subject only to the displayed valuation bounds. For example, with $p = 5$ and $m = 49$, the polynomial $(X^{49} - A)^5 + 7$ is irreducible for every integer A.

The extension is an immediate corollary of the previously established arithmetic exclusion and a standard binomial criterion. It does not add a new coefficient-integrality hypothesis or an assumed affine-exclusion callback to the final statement. The inherited arithmetic argument is in *Irreducibility of a family of shifted binomial compositions*; the original sealed source bundle is identified in Section 4.

2 Proof in the actual root field

Proof. Since $2 \leq p$, the condition on prime divisors implies $2 \nmid m$. Thus m is odd. The prime $p \geq 5$ is also odd.

The valuation hypotheses imply that k is not a pth power in $\mathbb{Q}$: any prime divisor ℓ of k has $0 < v_\ell(k) < p$, whereas a rational pth power has valuation divisible by p. The prime-binomial

criterion therefore makes $T^p - k$ irreducible over $\mathbb{Q}$. As p is odd, changing the sign of the variable also proves that

$$f(T) = T^p + k$$

is irreducible.

Let α be a root of f, and work in its actual simple root field $K = \mathbb{Q}(\alpha)$. Put $\theta = -\alpha$. Then $\theta^p = k$, $K = \mathbb{Q}(\theta)$ and $[K : \mathbb{Q}] = p$. Consequently every $\beta \in K$ admits a reduced representation

$$\beta = G(\theta), \qquad G \in \mathbb{Q}[T], \qquad \deg G < p.$$

For each prime $q \mid m$ we have $q > p$. Applying the inherited checked affine-power exclusion to this representation gives

$$\beta^q \neq A - \theta = A + \alpha \qquad \text{for every } \beta \in K. \tag{1}$$

This is the same signed arithmetic endpoint used by the original paper; its hypotheses are precisely the prime and valuation assumptions above.

The standard odd-binomial irreducibility criterion says that, for odd m, the conditions (1) for every prime divisor q of m imply the irreducibility of

$$X^m - (A + \alpha) \quad \text{over } K.$$

It applies to arbitrary odd m, including those with repeated prime factors. With $g(X) = X^m - A$, this binomial is exactly $g(X) - \alpha$. The monic composition criterion now gives the irreducibility over $\mathbb{Q}$ of

$$f(g(X)) = (X^m - A)^p + k,$$

as required. □

3 Exact theorem to Lean map

The newly compiled module is `CompositeInnerPillai`. Its exact root is

`PillaiScoped.irreducible_composition_of_inner_prime_factors_gt`.

The parameters p, m, q, ℓ have type $\mathbb{N}$, while k, A have type $\mathbb{Z}$. The root concludes literal polynomial irreducibility in $\mathbb{Q}[X]$. Its valuation premise is

$$\forall \ell \in \mathbb{N}, \quad \ell \text{ prime} \Longrightarrow (\ell : \mathbb{Z}) \mid k \Longrightarrow \text{padicValInt}(\ell, k) < p.$$

Since $|k| > 1$, the convention for the valuation of zero is irrelevant. There are no further mathematical premises.

1. **Outer binomial irreducibility.**
   `PillaiGlobalCoefficients.rational_binomial_irreducible_of_valuation_bounds`, followed by `PillaiCapelliBridge.outer_irreducible`.
2. **Reduced coordinates in K and the sign.**
   `PillaiCapelliBridge.exists_reduced_neg_generator_representation` and `PillaiCapelliBridge.neg_generator_power`.
3. **Exclusion for every prime $q \mid m$.** `PillaiScoped.reduced_affine_power_equation_exclusion`.
4. **Odd inner binomial and composition.** `X_pow_sub_C_irreducible_of_odd`, then `Polynomial.irreducible_comp` and `PillaiCapelliBridge.composition_identity`.

4 Verification and reproducibility record

Current verification boundary. The new root compiled with Lean 4.32.2. The recorded audit checks the root, the affine-power exclusion, the reduced-representation theorem, the valuation-based binomial theorem, and the odd-binomial criterion. All five reported exactly the standard axioms `propext`, `Classical.choice` and `Quot.sound`. An independent AI-assisted semantic rereview of the final compiled source and printed endpoint passed. This is not external expert peer review.

Independent full-closure checking of this new root passed. The production theorem type and referenced definitions matched a separately authored, reviewed requested-type specification. Its complete serialized closure passed replay in the patched official Lean 4.34.1 kernel and independent nanoda 0.4.19 checking, with 45,387 declarations. The 295,701,911-byte production export contains exactly the three standard axioms and no `sorryAx`. All 26 mandatory primitive definitions and the types of every permitted axiom present matched a separately generated trusted official Lean 4.32.2 baseline.

All 4,527 expected imported-object hashes matched. This is a source-completed expected inventory, not a fresh runtime module enumeration. The checks used a non-isolated validation adapter; they are not sandboxed comparator certification or external expert peer review. The separate unproved specification supplied only the requested type and was excluded from the production proof and its imports. Compact records, including the full production-export and result-file hashes, accompany this addendum.

Pinned source identities. The extension source is `CompositeInnerPillai.lean`, with SHA256 shown below as two consecutive blocks:

`4cf000fcd76a37ee0b77b269db503ca4 9b7b429c190be5f527e6bb009f8bd6f2`

The upstream odd-binomial criterion is in `Mathlib.FieldTheory.KummerExtension`, at Mathlib revision

`905b95818eb32af7874a58b427f50c1711a5e96c`.

The inherited prerequisite is the separate sealed archive `scoped-irreducibility-20261012-final.zip`, whose SHA256 is

`926619654aace3d34b6fc7f8fba3c014e bdfb1844e00573b1e2569a1d29b7822`.

The small companion source package contains this addendum's editable LaTeX and PDF build script, the exact extension and audit sources, the unchanged pinned upstream source with its license notices, and compact build, audit and semantic-review evidence. The original archive supplies the inherited local proof sources. The addendum package is not a standalone Lean dependency distribution. It includes no prebuilt proof objects, admitted type-comparison fixture, or serialized checker export.

Authorship and research status. This document was prepared with AI assistance. The user-selected project series label "Noah's GPT-6 Astra math problems" is not a certification of the identity or version of the model used. This addendum asserts no novelty or priority. Formal verification of the stated scoped result, originality, and the full fixed-gap Pillai conjecture are distinct questions; the full conjecture remains unresolved.